

## **Full Stack Engineer Technical Assessment**

**Project:** Build an AI-Ready Job Portal with Job Aggregation

**Duration:** 24–48 Hours

**Objective:** Evaluate frontend, backend, database design, REST APIs, authentication, web scraping, deployment, and code quality.

### **Recommended Tech Stack**

- Frontend: React/Next.js + Tailwind CSS
- Backend: Node.js (Express/NestJS) or Python (FastAPI)
- Database: PostgreSQL (preferred) / MySQL / MongoDB
- Authentication: JWT
- Deployment: Vercel/Render/Railway/AWS

### **Module 1 – Authentication**

- Candidate & Employer registration/login
- JWT authentication
- Forgot password (optional)
- Role-based access: Admin, Employer, Candidate

### **Module 2 – Job Portal**

- Employer: Create/Edit/Delete/Close jobs and view applicants
- Candidate: Search, filter, save and apply to jobs
- Job fields: title, company, location, work mode, employment type, salary, experience, skills, description, benefits, deadline

### **Module 3 – REST APIs**

- POST /auth/register
- POST /auth/login
- GET /jobs
- POST /jobs
- PUT /jobs/:id
- DELETE /jobs/:id
- GET /jobs/:id
- POST /jobs/:id/apply
- GET /applications
- GET /dashboard
- Support pagination, filtering, sorting and search

### **Module 4 – Database**

- Tables: Users, Roles, Companies, Jobs, Applications, SavedJobs

- Use proper relationships, constraints and indexing

### **Module 5 – Job Scraping**

- Scrape jobs from public sources that permit automated access (or use publicly available feeds/APIs).
- Store source, URL, company, title, location, description, skills, posted date.
- Prevent duplicate records.

### **Module 6 – Scraper API**

- Create POST /scrape/jobs endpoint.
- Return jobs added, duplicates skipped and errors.
- Bonus: schedule every 6 hours.

### **Module 7 – Admin Dashboard**

- Display total users, jobs, companies, applications, jobs scraped today, top skills, companies and locations.

### **Module 8 – Frontend**

- Responsive pages: Home, Login, Register, Job Listing, Job Details, Employer Dashboard, Candidate Dashboard, Admin Dashboard, Profile

### **Bonus Features (Choose Any Three)**

- Docker
- Swagger/OpenAPI
- Unit Tests
- Redis Cache
- CI/CD
- Resume Upload
- Email Notifications
- Rate Limiting
- AI Resume Matching API

### **Deliverables**

- GitHub Repository
- Live Deployment URL
- Database Schema
- API Documentation
- Postman Collection
- README

**Submission Guidelines:**

 Send your submission to: [build@innov2grow.in](mailto:build@innov2grow.in)

 Subject line: Name\_Role\_Task\_Submission

send the live URL + along with your resume

The solution should follow clean architecture, include proper error handling, validation, reusable components, environment configuration, and clear setup instructions.